Brutal Practice Paper B37 - Nutrition in Plants, Lines & Angles, Electric Current & its Effects, Large Numbers

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. During photosynthesis a banyan leaf makes glucose at a steady 2 grams every hour. If it gets 7 hours of bright sunlight, how much glucose forms and which gas is given out? [Nutrition in Plants]

- **(A)** 14 g of glucose, and carbon dioxide is given out
- **(B)** 21 g of glucose, and oxygen is given out
- **(C)** 7 g of glucose, and oxygen is given out
- **(D)** 9 g of glucose, and oxygen is given out
- **(E)** 14 g of glucose, and oxygen is given out
- **(F)** 2 g of glucose, and oxygen is given out
- **(G)** 14 g of starch, and oxygen is given out

2. An angle is 4 times its own complement. Find the angle, and then find its supplement. [Lines & Angles]

- **(A)** Angle = 60 degrees, supplement = 120 degrees
- **(B)** Angle = 72 degrees, supplement = 28 degrees
- **(C)** Angle = 72 degrees, supplement = 108 degrees
- **(D)** Angle = 36 degrees, supplement = 144 degrees
- **(E)** Angle = 80 degrees, supplement = 100 degrees
- **(F)** Angle = 72 degrees, supplement = 18 degrees
- **(G)** Angle = 18 degrees, supplement = 162 degrees

3. Four cells, each marked 1.5 V, are joined end to end (in series) to make a battery. What is the total voltage, and does series joining make a bulb brighter or dimmer than one cell? [Electric Current & its Effects]

- **(A)** 8 V; brighter
- **(B)** 6 V; same as one cell
- **(C)** 6 V; dimmer
- **(D)** 3 V; brighter
- **(E)** 6 V; brighter
- **(F)** 1.5 V; brighter

4. In the number 5,32,07,841 (Indian system), what is the place value of the digit 2, and what is its place called? [Large Numbers]

- **(A)** 20,000; the ten-thousands place
- **(B)** 2,00,000; the lakhs place
- **(C)** 200; the hundreds place
- **(D)** 2,000; the thousands place
- **(E)** 20,00,000; the lakhs place
- **(F)** 2,00,000; the ten-thousands place

5. A leaf has about 300 stomata in each square millimetre of its lower surface. A square patch measuring 2 mm by 2 mm is studied. How many pores does it have, and what mainly enters through them for food-making? [Nutrition in Plants]

- **(A)** 1200 pores; oxygen enters
- **(B)** 1200 pores; glucose enters
- **(C)** 900 pores; carbon dioxide enters
- **(D)** 1200 pores; water enters
- **(E)** 1200 pores; carbon dioxide enters
- **(F)** 300 pores; carbon dioxide enters
- **(G)** 600 pores; carbon dioxide enters

6. Two angles form a linear pair and are in the ratio 4 : 5. Find both angles and the difference between them. [Lines & Angles]

- **(A)** 75 degrees and 105 degrees, difference 30 degrees
- **(B)** 90 degrees and 100 degrees, difference 10 degrees
- **(C)** 72 degrees and 90 degrees, difference 18 degrees
- **(D)** 80 degrees and 100 degrees, difference 20 degrees
- **(E)** 40 degrees and 50 degrees, difference 10 degrees
- **(F)** 100 degrees and 80 degrees, difference 200 degrees
- **(G)** 81 degrees and 99 degrees, difference 18 degrees

7. Three identical bulbs are connected in series on one wire. One bulb's filament breaks. What happens to the other two, and why? [Electric Current & its Effects]

- **(A)** Both stay on, because each bulb has its own path
- **(B)** Both go off, because the battery dies
- **(C)** Only one of them goes off
- **(D)** Both stay on but dimmer, because of the gap
- **(E)** Both go off, because the broken bulb breaks the single path for current
- **(F)** Both glow, because current jumps the gap
- **(G)** Both glow brighter, because more current flows

8. Write 73525410 with commas in the International system, and state how many millions it has. [Large Numbers]

- **(A)** 735,254,10; it has 73 millions
- **(B)** 73,525,410; it has 7 millions
- **(C)** 7,35,25,410; it has 73 millions
- **(D)** 73,525,410; it has 735 millions
- **(E)** 73,525,410; it has 73 millions
- **(F)** 73,525,410; it has 3 millions

9. A variegated leaf has a total surface of 240 cm2, and 5/8 of it is green. After the iodine starch test, what area turns blue-black, and why only that part? [Nutrition in Plants]

- **(A)** 120 cm2 turns blue-black, because half the leaf is green
- **(B)** 90 cm2 turns blue-black, because only the green part makes starch
- **(C)** 150 cm2 turns red, because chlorophyll makes starch
- **(D)** 150 cm2 turns blue-black, because only the green chlorophyll part makes starch
- **(E)** 240 cm2 turns blue-black, because the whole leaf makes starch
- **(F)** 48 cm2 turns blue-black, because only green makes starch
- **(G)** 150 cm2 turns blue-black, because the white part stores starch

10. Two straight lines cross each other. One of the four angles is 110 degrees. What are the measures of the angle vertically opposite to it and the angle next to it (its linear pair)? [Lines & Angles]

- **(A)** Vertically opposite = 70 degrees, adjacent = 70 degrees
- **(B)** Vertically opposite = 110 degrees, adjacent = 110 degrees
- **(C)** Vertically opposite = 90 degrees, adjacent = 70 degrees
- **(D)** Vertically opposite = 250 degrees, adjacent = 70 degrees
- **(E)** Vertically opposite = 110 degrees, adjacent = 90 degrees
- **(F)** Vertically opposite = 70 degrees, adjacent = 110 degrees
- **(G)** Vertically opposite = 110 degrees, adjacent = 70 degrees

11. An electric heater uses a coil that glows red-hot when current passes - the heating effect of current. If the heater runs 3 hours a day for 30 days, how many hours is that, and the coil is usually made of? [Electric Current & its Effects]

- **(A)** 90 hours; the coil is made of nichrome
- **(B)** 90 hours; the coil is made of silver
- **(C)** 60 hours; the coil is made of nichrome
- **(D)** 90 hours; the coil is made of copper
- **(E)** 90 hours; the coil is made of plastic
- **(F)** 33 hours; the coil is made of nichrome
- **(G)** 900 hours; the coil is made of nichrome

12. Round 4,57,832 to the nearest thousand, then round that result to the nearest ten-thousand. [Large Numbers]

- **(A)** 4,60,000, then 4,60,000
- **(B)** 4,58,000, then 5,00,000
- **(C)** 4,58,000, then 4,60,000
- **(D)** 4,58,000, then 4,58,000
- **(E)** 4,57,000, then 4,60,000
- **(F)** 4,57,800, then 4,60,000
- **(G)** 4,58,000, then 4,50,000

13. A pea field needs 60 kg of nitrogen. Helpful bacteria in the root nodules supply 3/5 of it free, and the farmer adds the rest as fertiliser. How much fertiliser-nitrogen is needed, and what is the bacterium called? [Nutrition in Plants]

- **(A)** 12 kg, and the bacterium is Rhizobium
- **(B)** 36 kg, and the bacterium is Rhizobium
- **(C)** 24 kg, and the bacterium is Rhizobium
- **(D)** 24 kg, and the bacterium is Cuscuta
- **(E)** 24 kg, and the bacterium is chlorophyll
- **(F)** 20 kg, and the bacterium is Rhizobium
- **(G)** 36 kg, and the bacterium is a lichen

14. A transversal cuts two parallel lines. A pair of co-interior (same-side interior) angles are x and 2x. Find x and the larger angle. [Lines & Angles]

(A) x = 90 degrees, larger angle = 180 degrees
(B) x = 60 degrees, larger angle = 120 degrees
(C) x = 45 degrees, larger angle = 90 degrees
(D) x = 120 degrees, larger angle = 240 degrees
(E) x = 60 degrees, larger angle = 100 degrees
(F) x = 60 degrees, larger angle = 60 degrees
(G) x = 30 degrees, larger angle = 60 degrees

15. A circuit normally carries a current of 4 A. Which fuse is the sensible choice, and what does a fuse do when too much current flows? [Electric Current & its Effects]

(A) A 5 A fuse; it cools the wires
(B) A 5 A fuse; it melts and breaks the circuit when current is too high
(C) A 5 A fuse; it makes the current stronger
(D) A 3 A fuse; it melts when current is too high
(E) A 2 A fuse; it breaks the circuit on overload
(F) A 5 A fuse; it stores current for later
(G) A 10 A fuse that never melts

16. Which of these is the correct expanded form of 60,508? [Large Numbers]

(A) 6,000 + 50 + 8
(B) 60,000 + 500 + 8
(C) 60,000 + 500 + 80
(D) 60,000 + 508
(E) 60,000 + 580
(F) 6,000 + 500 + 8
(G) 60,000 + 5,000 + 8

17. Cuscuta (amarbel) climbs over a host and takes ready-made food from it. If a host loses 15% of its 800 g of stored food to the amarbel in a week, how much food is left, and what type of nutrition does amarbel show? [Nutrition in Plants]

(A) 680 g left; saprotrophic nutrition
(B) 680 g left; autotrophic nutrition
(C) 685 g left; parasitic nutrition
(D) 640 g left; parasitic nutrition
(E) 680 g left; parasitic nutrition
(F) 720 g left; parasitic nutrition
(G) 120 g left; parasitic nutrition

18. A transversal crosses two parallel lines. An angle and its corresponding angle are 5y and 65 degrees. Find y, then the angle that makes a linear pair with the 65 degree angle. [Lines & Angles]

(A) y = 13, linear pair angle = 90 degrees
(B) y = 13, linear pair angle = 115 degrees
(C) y = 325, linear pair angle = 115 degrees
(D) y = 13, linear pair angle = 35 degrees
(E) y = 65, linear pair angle = 115 degrees
(F) y = 12, linear pair angle = 115 degrees
(G) y = 13, linear pair angle = 25 degrees

19. A student winds 50 turns of wire on an iron nail to make an electromagnet, lifting 10 pins. She rewinds it with 150 turns. About how many pins might it now lift, and why? [Electric Current & its Effects]

- **(A)** About 10 pins, because the number of turns does not matter
- **(B)** About 30 pins, because more turns make the electromagnet stronger
- **(C)** About 150 pins, because pins equal the turns
- **(D)** About 5 pins, because more turns weaken it
- **(E)** About 30 pins, because more turns make it weaker
- **(F)** About 30 pins, because the iron melts

20. A factory made 12,50,000 toys and sold 8,76,540 of them. How many toys are left? [Large Numbers]

- **(A)** 3,72,560
- **(B)** 3,73,540
- **(C)** 2,73,460
- **(D)** 21,26,540
- **(E)** 3,73,460
- **(F)** 4,73,460
- **(G)** 3,74,460

21. A pitcher plant traps insects to get the one nutrient it cannot get from its boggy soil. If it digests 8 insects a day for 5 days, that is how many insects, and which nutrient is it mainly after? [Nutrition in Plants]

- **(A)** 40 insects; glucose
- **(B)** 40 insects; nitrogen
- **(C)** 45 insects; nitrogen
- **(D)** 32 insects; nitrogen
- **(E)** 40 insects; oxygen
- **(F)** 40 insects; carbon
- **(G)** 13 insects; nitrogen

22. Three angles meet at a point and together go all the way round. Two of them are 120 degrees and 150 degrees. Find the third angle and say what type it is. [Lines & Angles]

- **(A)** 90 degrees, an obtuse angle
- **(B)** 90 degrees, a straight angle
- **(C)** 30 degrees, a right angle
- **(D)** 60 degrees, a right angle
- **(E)** 90 degrees, a right angle
- **(F)** 100 degrees, a right angle
- **(G)** 90 degrees, an acute angle

23. In a circuit diagram a cell is drawn as a long thin line beside a short thick line. Which line is the positive terminal, and what do two or more cells together form? [Electric Current & its Effects]

- **(A)** The long line is negative; together they form a battery
- **(B)** Both lines are positive; together they form a battery
- **(C)** The long line is positive; together they form a battery
- **(D)** The long line is positive; together they form a bulb
- **(E)** The short thick line is positive; together they form a switch
- **(F)** The long line is positive; together they form a fuse
- **(G)** The short line is positive; together they form a battery

24. Estimate the product 412 x 589 by rounding each number to the nearest hundred. What estimate do you get, and is the exact answer a little more or a little less? [Large Numbers]

- **(A)** 250000; the exact answer is a little more
- **(B)** 240000; the exact answer is equal
- **(C)** 230000; the exact answer is a little more
- **(D)** 210000; the exact answer is a little more
- **(E)** 240000; the exact answer is a little more
- **(F)** 240000; the exact answer is a little less
- **(G)** 2400000; the exact answer is a little less

25. A saprotrophic fungus on a slice of bread covers 2 cm2 today and doubles the area it covers every day. What area will it cover after 4 days, and how does it get its food? [Nutrition in Plants]

- **(A)** 64 cm2; by absorbing the digested food
- **(B)** 32 cm2; by making its own food using sunlight
- **(C)** 10 cm2; by absorbing the digested food
- **(D)** 16 cm2; by releasing juices and absorbing the digested food
- **(E)** 32 cm2; by trapping insects
- **(F)** 32 cm2; by releasing juices onto the dead bread and absorbing the digested food
- **(G)** 8 cm2; by absorbing the digested food

26. Two angles (2x + 10) degrees and (3x - 30) degrees form a linear pair. Find x and the two angles. [Lines & Angles]

- **(A)** x = 36, angles 82 degrees and 78 degrees
- **(B)** x = 40, angles 90 degrees and 80 degrees
- **(C)** x = 30, angles 70 degrees and 60 degrees
- **(D)** x = 40, angles 80 degrees and 100 degrees
- **(E)** x = 40, angles 90 degrees and 90 degrees
- **(F)** x = 44, angles 98 degrees and 102 degrees
- **(G)** x = 40, angles 100 degrees and 80 degrees

27. Two identical bulbs are run from the same battery, once in series and once in parallel. In which arrangement is each bulb brighter, and why? [Electric Current & its Effects]

- **(A)** Both arrangements give equal brightness
- **(B)** In parallel, because the current halves
- **(C)** In series, because current passes through both
- **(D)** In series, because the voltage adds up
- **(E)** In parallel, because the bulbs share one filament
- **(F)** In series, because each gets the full voltage
- **(G)** In parallel, because each bulb gets the full battery voltage

28. Write the predecessor and the successor of 7,00,000. [Large Numbers]

- **(A)** Predecessor 6,99,000; successor 7,01,000
- **(B)** Predecessor 7,00,001; successor 6,99,999
- **(C)** Predecessor 6,99,999; successor 7,00,011
- **(D)** Predecessor 6,99,999; successor 7,00,000
- **(E)** Predecessor 6,99,990; successor 7,00,010
- **(F)** Predecessor 6,99,999; successor 7,00,001

29. In a lichen an alga and a fungus live together and help each other to survive. What is this kind of living-together called, and which partner actually makes the food? [Nutrition in Plants]

- **(A)** Parasitism; the alga makes the food
- **(B)** Symbiosis; the alga makes the food
- **(C)** Saprotrophism; the alga makes the food
- **(D)** Insectivory; the alga makes the food
- **(E)** Symbiosis; the fungus makes food using sunlight
- **(F)** Symbiosis; both make food by photosynthesis

30. The supplement of an angle is 4 times the angle itself. Find the angle and its complement. [Lines & Angles]

- **(A)** Angle = 36 degrees, complement = 54 degrees
- **(B)** Angle = 72 degrees, complement = 18 degrees
- **(C)** Angle = 45 degrees, complement = 45 degrees
- **(D)** Angle = 144 degrees, complement = 54 degrees
- **(E)** Angle = 36 degrees, complement = 64 degrees
- **(F)** Angle = 36 degrees, complement = 144 degrees
- **(G)** Angle = 30 degrees, complement = 60 degrees

31. A compass needle near a current-carrying wire deflects to the right. When the battery's terminals are swapped, what happens to the needle, and what does this prove? [Electric Current & its Effects]

- **(A)** It deflects the other way (left), showing current produces a magnetic effect
- **(B)** It deflects the other way, showing only a chemical effect
- **(C)** It spins fully around, showing current makes light
- **(D)** It melts, showing the heating effect
- **(E)** It deflects further right, showing the heating effect
- **(F)** It points straight down, showing gravity
- **(G)** It stops moving, showing current has no magnetic effect

32. Which of these is the greatest number: 9,09,990; 9,90,099; 9,09,909; 9,90,090? [Large Numbers]

- **(A)** 9,90,090
- **(B)** they are all equal
- **(C)** 9,99,000
- **(D)** 9,09,990
- **(E)** 9,90,099
- **(F)** 9,09,909
- **(G)** 9,00,999

33. After growing wheat that uses up nitrogen, a farmer grows gram (a legume) to put nitrogen back. If gram restores 9 kg of nitrogen per 100 m2 and the field is 500 m2, how much is restored, and what is this practice called? [Nutrition in Plants]

- **(A)** 45 kg; saprotrophism
- **(B)** 40 kg; crop rotation
- **(C)** 45 kg; crop rotation
- **(D)** 45 kg; transpiration
- **(E)** 9 kg; crop rotation
- **(F)** 54 kg; crop rotation
- **(G)** 45 kg; photosynthesis

34. A transversal meets two parallel lines so that a pair of alternate interior angles are (4x - 5) degrees and (3x + 15) degrees. Find x and the size of each of these equal angles. [Lines & Angles]

- **(A)** x = 20, each angle = 75 degrees
- **(B)** x = 10, each angle = 35 degrees
- **(C)** x = 5, each angle = 15 degrees
- **(D)** x = 20, each angle = 80 degrees
- **(E)** x = 20, each angle = 85 degrees
- **(F)** x = 25, each angle = 95 degrees

35. Three 1.5 V cells are joined in series, but one is accidentally put in backwards. What is the net voltage of this battery? [Electric Current & its Effects]

- **(A)** 2.5 V
- **(B)** 0 V
- **(C)** 6 V
- **(D)** 1 V
- **(E)** 4.5 V
- **(F)** 3 V
- **(G)** 1.5 V

36. Using the digits 3, 0, 7, 5, 1 once each, form the greatest and the smallest 5-digit numbers (no leading zero for the smallest). [Large Numbers]

- **(A)** Greatest 75,310; smallest 13,057
- **(B)** Greatest 73,510; smallest 10,357
- **(C)** Greatest 75,301; smallest 10,357
- **(D)** Greatest 75,310; smallest 10,357
- **(E)** Greatest 75,310; smallest 01,357
- **(F)** Greatest 57,310; smallest 10,357

37. From this list - mango tree, mushroom, grass, cow, Cuscuta - how many are autotrophs (make their own food), and name one heterotroph from the list? [Nutrition in Plants]

- **(A)** 2 autotrophs; cow is a heterotroph
- **(B)** 2 autotrophs; mushroom is an autotroph
- **(C)** 3 autotrophs; cow is a heterotroph
- **(D)** 1 autotroph; cow is a heterotroph
- **(E)** 2 autotrophs; mango tree is a heterotroph
- **(F)** 2 autotrophs; grass is a heterotroph
- **(G)** 4 autotrophs; cow is a heterotroph

38. An angle of 80 degrees is cut into two equal parts by a ray. Find one part and the complement of that part. [Lines & Angles]

- **(A)** Each part 20 degrees, complement 70 degrees
- **(B)** Each part 80 degrees, complement 10 degrees
- **(C)** Each part 40 degrees, complement 60 degrees
- **(D)** Each part 160 degrees, complement 50 degrees
- **(E)** Each part 40 degrees, complement 140 degrees
- **(F)** Each part 40 degrees, complement 50 degrees
- **(G)** Each part 40 degrees, complement 10 degrees

39. Modern houses use an MCB instead of a fuse wire. What does an MCB do during a sudden surge of current, and what is its key advantage over a fuse? [Electric Current & its Effects]

- **(A)** It cools the wires, and it can be reset
- **(B)** It switches the circuit off, but it must be thrown away each time
- **(C)** It makes the bulbs brighter, and it can be reset
- **(D)** It increases the current, and it is cheaper
- **(E)** It melts like a wire, and it cannot be reset
- **(F)** It switches the circuit off, and unlike a fuse it can be reset and reused
- **(G)** It stores the extra current, and it can be reset

40. A company's profit is 2 crore rupees. How many lakhs is that, and how many zeros are there in 1 crore? [Large Numbers]

- **(A)** 2000 lakh; 1 crore has 7 zeros
- **(B)** 200 lakh; 1 crore has 5 zeros
- **(C)** 100 lakh; 1 crore has 7 zeros
- **(D)** 200 lakh; 1 crore has 8 zeros
- **(E)** 20 lakh; 1 crore has 7 zeros
- **(F)** 200 lakh; 1 crore has 6 zeros
- **(G)** 200 lakh; 1 crore has 7 zeros

41. A plant is kept in the dark for 2 full days before a starch test so its leaves use up stored starch. For how many hours is it in the dark, and why is this destarching step done? [Nutrition in Plants]

- **(A)** 12 hours; so that starch is made
- **(B)** 48 hours; so the leaf can absorb oxygen
- **(C)** 24 hours; so that the stored starch is used up
- **(D)** 72 hours; so the starch is removed
- **(E)** 48 hours; so the leaf turns green
- **(F)** 48 hours; so that any starch found in the test was made during the experiment
- **(G)** 48 hours; so the chlorophyll is destroyed

42. Each hour mark on a clock face is 30 degrees from the next. What is the angle between the two hands at exactly 4 o'clock, and what type of angle is it? [Lines & Angles]

- **(A)** 150 degrees, an obtuse angle
- **(B)** 120 degrees, an acute angle
- **(C)** 120 degrees, an obtuse angle
- **(D)** 100 degrees, an obtuse angle
- **(E)** 60 degrees, an obtuse angle
- **(F)** 120 degrees, a reflex angle
- **(G)** 90 degrees, a right angle

43. Two identical heaters carry the same current. Heater P runs for 20 minutes and heater Q for 35 minutes. Which gives out more heat, and the heating effect is used in which device - a torch bulb or a fan? [Electric Current & its Effects]

- **(A)** Heater Q gives more heat; a buzzer uses the heating effect
- **(B)** Heater P gives more heat; a torch bulb uses the heating effect
- **(C)** Heater P gives more heat; a fan uses the heating effect
- **(D)** Heater Q gives more heat; a torch bulb uses the heating effect
- **(E)** Both give equal heat; a fan uses the heating effect
- **(F)** Both give equal heat; a torch bulb uses the heating effect
- **(G)** Heater Q gives more heat; a fan uses the heating effect

44. A book has 250 pages. Each page has 40 lines, and each line has about 12 words. Roughly how many words are in the whole book? [Large Numbers]

- **(A)** 1,20,000 words
- **(B)** 12,000 words
- **(C)** 1,02,000 words
- **(D)** 1,00,000 words
- **(E)** 1,20,00,000 words
- **(F)** 12,00,000 words
- **(G)** 10,000 words

45. A tree has 5000 leaves, and each leaf makes 0.4 g of glucose on a sunny day. How much glucose does the whole tree make that day, and where is the extra kept? [Nutrition in Plants]

- **(A)** 2000 g; the extra is stored as oxygen in the leaves
- **(B)** 2000 g (2 kg); the extra is stored as starch in parts like roots and stem
- **(C)** 1250 g; the extra is stored as starch
- **(D)** 5000 g; the extra is stored as starch
- **(E)** 200 g; the extra is stored as starch
- **(F)** 2000 g; the extra is stored as glucose gas
- **(G)** 2 g; the extra is stored as starch

46. Two complementary angles differ by 24 degrees. Find both angles. [Lines & Angles]

- **(A)** 33 degrees and 57 degrees
- **(B)** 20 degrees and 44 degrees
- **(C)** 33 degrees and 47 degrees
- **(D)** 44 degrees and 68 degrees
- **(E)** 36 degrees and 60 degrees
- **(F)** 66 degrees and 24 degrees

47. From this list - copper, rubber, iron, plastic, aluminium - how many are good conductors of electricity, and name one insulator from the list? [Electric Current & its Effects]

- **(A)** 3 conductors; iron is an insulator
- **(B)** 4 conductors; plastic is an insulator
- **(C)** 5 conductors; rubber is an insulator
- **(D)** 3 conductors; copper is an insulator
- **(E)** 3 conductors; rubber is an insulator
- **(F)** 1 conductor; rubber is an insulator
- **(G)** 2 conductors; rubber is an insulator

48. Estimate 3,46,789 + 2,53,211 by rounding each number to the nearest hundred, then give the exact sum. [Large Numbers]

- **(A)** Estimate 6,00,000; exact 5,99,000
- **(B)** Estimate 7,00,000; exact 6,00,000
- **(C)** Estimate 5,00,000; exact 6,00,000
- **(D)** Estimate 6,00,000; exact 6,99,000
- **(E)** Estimate 6,00,000; exact 6,00,000
- **(F)** Estimate 6,00,000; exact 6,00,100
- **(G)** Estimate 6,10,000; exact 6,00,000

49. On a bright afternoon a leaf takes in 90 mg of one gas and gives out 110 mg of another through its stomata for photosynthesis. Which gas is taken in, and what is the difference between the two masses? [Nutrition in Plants]

- **(A)** Oxygen is taken in; the difference is 20 mg
- **(B)** Carbon dioxide is taken in; the difference is 200 mg
- **(C)** Carbon dioxide is taken in; the difference is 10 mg
- **(D)** Carbon dioxide is taken in; the difference is 20 mg
- **(E)** Nitrogen is taken in; the difference is 20 mg
- **(F)** Carbon dioxide is taken in; the difference is 90 mg
- **(G)** Oxygen is taken in; the difference is 200 mg

50. Two parallel roads are crossed by a third straight road. One of the eight angles formed is 75 degrees. Find the co-interior angle paired with it. [Lines & Angles]

- **(A)** 115 degrees
- **(B)** 15 degrees
- **(C)** 285 degrees
- **(D)** 25 degrees
- **(E)** 75 degrees
- **(F)** 105 degrees
- **(G)** 95 degrees

51. A scrapyard crane uses an electromagnet, not a permanent magnet, to pick up and drop iron pieces. Why an electromagnet, and what happens when the current is switched off? [Electric Current & its Effects]

- **(A)** Because it lifts plastic; switching off drops the iron
- **(B)** Because it is cheaper; the iron stays stuck forever
- **(C)** Because it never loses magnetism; it keeps holding the iron
- **(D)** Because it can be turned on and off; switching off makes it lose magnetism and drop the iron
- **(E)** Because it can be turned on and off; switching off makes it stronger
- **(F)** Because it makes heat; the iron melts off
- **(G)** Because it is a permanent magnet; it drops the iron

52. Divide 9,99,999 by 7. What is the quotient and the remainder? [Large Numbers]

- **(A)** Quotient 1,42,857; remainder 0
- **(B)** Quotient 1,42,758; remainder 0
- **(C)** Quotient 1,42,867; remainder 0
- **(D)** Quotient 1,42,857; remainder 7
- **(E)** Quotient 1,42,857; remainder 1
- **(F)** Quotient 1,43,857; remainder 0
- **(G)** Quotient 1,42,850; remainder 0

53. Six leaves are tested with iodine; 4 turn blue-black and 2 stay brown. What fraction of the leaves had made starch, and what does the brown colour tell you? [Nutrition in Plants]

- **(A)** 2/3 of the leaves; brown means oxygen was made there
- **(B)** 1/2 of the leaves; brown means no starch was made there
- **(C)** 4/6 of the leaves; brown means starch was made there
- **(D)** 2/3 of the leaves; brown means no starch was made there
- **(E)** 2/3 of the leaves; brown means extra starch was made there
- **(F)** 3/5 of the leaves; brown means no starch was made there

54. An angle measures 100 degrees. What is its reflex angle, and is the original angle acute, right, or obtuse? [Lines & Angles]

- **(A)** 260 degrees, and the angle is right
- **(B)** 260 degrees, and the angle is acute
- **(C)** 260 degrees, and the angle is obtuse
- **(D)** 160 degrees, and the angle is obtuse
- **(E)** 270 degrees, and the angle is obtuse
- **(F)** 80 degrees, and the angle is obtuse

55. A torch needs 2 cells to work. A shopkeeper has 6 cells in stock. How many full sets can he supply, and joining the 2 cells in series gives what voltage if each is 1.5 V? [Electric Current & its Effects]

- **(A)** 4 sets; 3 V
- **(B)** 3 sets; 3 V
- **(C)** 3 sets; 1.5 V
- **(D)** 3 sets; 6 V
- **(E)** 3 sets; 4.5 V
- **(F)** 2 sets; 3 V

56. The Moon is about 3,84,000 km from Earth. A spacecraft travels there and back. What total distance does it cover? [Large Numbers]

- **(A)** 7,68,000 km
- **(B)** 7,68,00 km
- **(C)** 11,52,000 km
- **(D)** 3,84,000 km
- **(E)** 3,68,000 km
- **(F)** 6,68,000 km
- **(G)** 7,86,000 km

57. In an experiment a leaf is sealed in a flask with potassium hydroxide, which soaks up all the carbon dioxide. After a sunny day the leaf is tested for starch. What result is expected, and why? [Nutrition in Plants]

- **(A)** No colour change, because iodine needs oxygen
- **(B)** No blue-black colour, because the leaf had no sunlight
- **(C)** Blue-black all over, because potassium hydroxide adds carbon
- **(D)** No blue-black colour, because without carbon dioxide the leaf cannot make starch
- **(E)** Blue-black all over, because sunlight alone makes starch
- **(F)** Red colour, because carbon dioxide was removed
- **(G)** No blue-black colour, because the chlorophyll was removed

58. Three angles lie along a straight line and are in the ratio 1 : 2 : 3. Find the angles and the type of the largest one. [Lines & Angles]

- **(A)** 30, 60, 90 degrees; the largest is acute
- **(B)** 30, 60, 90 degrees; the largest is a right angle
- **(C)** 45, 60, 75 degrees; the largest is obtuse
- **(D)** 60, 120, 180 degrees; the largest is straight
- **(E)** 20, 40, 60 degrees; the largest is acute
- **(F)** 30, 50, 100 degrees; the largest is right
- **(G)** 30, 60, 90 degrees; the largest is obtuse

59. A circuit has a battery, a bulb, and a switch. When the switch is open (off), is there a complete path, and does the bulb glow? [Electric Current & its Effects]

- **(A)** A complete path; the bulb glows brighter
- **(B)** A complete path; the bulb does not glow
- **(C)** No path; the battery glows instead
- **(D)** No path; the bulb glows dimly
- **(E)** A complete path; the bulb glows
- **(F)** No complete path; the bulb does not glow
- **(G)** No complete path; the bulb still glows

60. In the number 8,84,521, find the difference between the place values of the two 8s. [Large Numbers]

- **(A)** 8,00,080
- **(B)** 8,80,000
- **(C)** 0
- **(D)** 16,000
- **(E)** 72,000
- **(F)** 7,20,000
- **(G)** 7,02,000

61. Leaf A makes 8 g of glucose and Leaf B makes 10 g on the same day. By what percentage does Leaf B beat Leaf A, and which raw material from the air did both need? [Nutrition in Plants]

- **(A)** 2% more; both needed carbon dioxide
- **(B)** 12.5% more; both needed carbon dioxide
- **(C)** 18% more; both needed carbon dioxide
- **(D)** 20% more; both needed carbon dioxide
- **(E)** 25% more; both needed nitrogen
- **(F)** 25% more; both needed carbon dioxide
- **(G)** 25% more; both needed oxygen

62. Two lines cross. Two adjacent angles are (5x) degrees and (4x) degrees. Find x, then the angle vertically opposite to the (5x) degree angle. [Lines & Angles]

- **(A)** x = 20, vertically opposite angle = 80 degrees
- **(B)** x = 10, vertically opposite angle = 50 degrees
- **(C)** x = 20, vertically opposite angle = 90 degrees
- **(D)** x = 18, vertically opposite angle = 90 degrees
- **(E)** x = 20, vertically opposite angle = 260 degrees
- **(F)** x = 20, vertically opposite angle = 100 degrees
- **(G)** x = 22, vertically opposite angle = 110 degrees

63. The wire used in a heater coil must get very hot without melting. Which property must it have, and copper is NOT used for the coil because it - [Electric Current & its Effects]

- **(A)** A low melting point; copper is too expensive
- **(B)** A bright colour; copper is too dull
- **(C)** A high melting point; copper is too good a conductor and would not heat up enough
- **(D)** A low melting point; copper melts too slowly
- **(E)** Be very flexible; copper is too soft
- **(F)** A high melting point; copper is magnetic

64. Write 40,05,007 in words (Indian system). [Large Numbers]

- **(A)** Forty lakh five thousand seven
- **(B)** Forty lakh fifty thousand seven
- **(C)** Forty lakh five hundred seven
- **(D)** Forty lakh five thousand seventy
- **(E)** Four lakh five thousand seven
- **(F)** Four million five thousand seven
- **(G)** Forty-five lakh seven

65. A legume root has 40 nodules, and each nodule fixes 5 mg of nitrogen a day. How much nitrogen is fixed in a day, and the gas they convert comes from where? [Nutrition in Plants]

- **(A)** 2000 mg; the nitrogen comes from the air
- **(B)** 200 mg; the nitrogen gas comes from the air
- **(C)** 200 mg; the nitrogen comes from the water
- **(D)** 45 mg; the nitrogen comes from the air
- **(E)** 200 mg; the carbon dioxide comes from the air
- **(F)** 20 mg; the nitrogen comes from the air
- **(G)** 200 mg; the oxygen comes from the air

66. The complement of an angle is exactly one-third of its supplement. Find the angle. [Lines & Angles]

- **(A)** 54 degrees
- **(B)** 30 degrees
- **(C)** 45 degrees
- **(D)** 90 degrees
- **(E)** 75 degrees
- **(F)** 36 degrees
- **(G)** 60 degrees

67. By convention, in the wires outside the cell, current is taken to flow from which terminal to which, and electrons actually flow the opposite way - true or false? [Electric Current & its Effects]

- **(A)** Both ways at once; the statement is true
- **(B)** From negative to positive; and the statement is true
- **(C)** From positive to negative; and yes, that statement is true
- **(D)** From positive to positive; the statement is true
- **(E)** From negative to positive; and the statement is false
- **(F)** From positive to negative; and the statement is false
- **(G)** It does not flow at all; the statement is false

68. Estimate 198 x 102 by rounding each number to the nearest hundred, then say whether the real product is bigger or smaller than the estimate. [Large Numbers]

- **(A)** 20,000; the real product is smaller
- **(B)** 10,000; the real product is smaller
- **(C)** 20,000; the real product is bigger
- **(D)** 19,000; the real product is bigger
- **(E)** 20,000; they are exactly equal
- **(F)** 30,000; the real product is bigger
- **(G)** 2,00,000; the real product is bigger

69. Photosynthesis happens inside tiny green bodies in the leaf cells. If one cell has 50 of these and a tissue has 600 cells, how many are there in all, and what are these green bodies called? [Nutrition in Plants]

- **(A)** 30,000; they are called chlorophyll
- **(B)** 650; they are called chloroplasts
- **(C)** 30,000; they are called stomata
- **(D)** 30,000; they are called chloroplasts
- **(E)** 30,000; they are called nodules
- **(F)** 12,000; they are called chloroplasts
- **(G)** 3000; they are called chloroplasts

70. On two parallel lines a transversal makes a 'Z' shape with a pair of equal angles. If one of them is 58 degrees, what is the other, and what are such angles called? [Lines & Angles]

- **(A)** 58 degrees; they are co-interior angles
- **(B)** 32 degrees; they are alternate interior angles
- **(C)** 58 degrees; they are corresponding angles
- **(D)** 122 degrees; they are alternate interior angles
- **(E)** 58 degrees; they are linear-pair angles
- **(F)** 58 degrees; they are alternate interior angles
- **(G)** 90 degrees; they are alternate interior angles

71. One bulb on a battery glows brightly. Two more identical bulbs are then added in series. What happens to the brightness of the first bulb, and why? [Electric Current & its Effects]

- **(A)** It flickers, because the current reverses
- **(B)** It stays the same, because the bulbs are identical
- **(C)** It goes off completely, because of the extra bulbs
- **(D)** It becomes dimmer, because the battery gains voltage
- **(E)** It becomes brighter, because the resistance drops
- **(F)** It becomes brighter, because more bulbs add power
- **(G)** It becomes dimmer, because the battery's push is now shared among three bulbs

72. A shop has 1,250 boxes of pens with 36 pens in each box. If each pen sells for 8 rupees, what is the total selling price of all the pens? [Large Numbers]

- **(A)** Rs 3,60,000
- **(B)** Rs 2,88,000
- **(C)** Rs 3,06,000
- **(D)** Rs 4,50,000
- **(E)** Rs 3,60,00
- **(F)** Rs 45,000

73. A water plant gives off 12 oxygen bubbles a minute in bright light. How many bubbles in 15 minutes, and what does counting bubbles measure? [Nutrition in Plants]

- **(A)** 1215 bubbles; it measures the photosynthesis rate
- **(B)** 180 bubbles; it measures how fast respiration is going
- **(C)** 27 bubbles; it measures the photosynthesis rate
- **(D)** 180 bubbles; it measures how much carbon dioxide is made
- **(E)** 160 bubbles; it measures the photosynthesis rate
- **(F)** 180 bubbles; it measures the water absorbed
- **(G)** 180 bubbles; it measures how fast photosynthesis is going

74. Ray OB lies between rays OA and OC. If angle AOC = 90 degrees and angle AOB = 35 degrees, find angle BOC and say whether AOB and BOC are complementary. [Lines & Angles]

- **(A)** 55 degrees; yes, they are supplementary
- **(B)** 55 degrees; no, they are a linear pair
- **(C)** 55 degrees; yes, they are complementary
- **(D)** 55 degrees; no, they are supplementary
- **(E)** 45 degrees; yes, they are complementary
- **(F)** 125 degrees; yes, they are complementary
- **(G)** 65 degrees; yes, they are complementary

75. An electric bell rings because an electromagnet pulls an iron strip, which breaks the circuit, springs back, and repeats. This rapid on-off action uses which effect of current? [Electric Current & its Effects]

- **(A)** The sound effect of current alone
- **(B)** The magnetic effect of current
- **(C)** The cooling effect of current
- **(D)** The heating effect of current
- **(E)** The effect of gravity
- **(F)** The lighting effect of current
- **(G)** The chemical effect of current

76. Which is larger: one million or ten lakh, and how many lakhs make one million? [Large Numbers]

- **(A)** They are equal; 100 lakh make a million
- **(B)** They are equal; ten lakh make one million
- **(C)** One million is larger; 100 lakh make a million
- **(D)** They are equal; 5 lakh make a million
- **(E)** Ten lakh is larger; 10 lakh make a million
- **(F)** One million is larger; 1 lakh makes a million
- **(G)** Ten lakh is larger; 1000 lakh make a million

77. A green plant and a mushroom both grow on a log. Which one is a producer, and if the plant makes 16 units of food while the mushroom makes 0, what is the mushroom's food source? [Nutrition in Plants]

- **(A)** The green plant is the producer; the mushroom eats the plant
- **(B)** The green plant is the producer; the mushroom absorbs food from the dead log
- **(C)** The mushroom is the producer; it makes 16 units too
- **(D)** Both are producers; the mushroom uses chlorophyll
- **(E)** The green plant is the producer; the mushroom traps insects
- **(F)** The green plant is the producer; the mushroom makes food from sunlight
- **(G)** The mushroom is the producer; the plant absorbs food from the log

78. The supplement of an angle is 30 degrees more than twice its complement. Find the angle and its supplement. [Lines & Angles]

- **(A)** Angle = 30 degrees, supplement = 90 degrees
- **(B)** Angle = 20 degrees, supplement = 160 degrees
- **(C)** Angle = 150 degrees, supplement = 30 degrees
- **(D)** Angle = 60 degrees, supplement = 120 degrees
- **(E)** Angle = 30 degrees, supplement = 150 degrees
- **(F)** Angle = 30 degrees, supplement = 60 degrees

79. A device needs about 9 V to run. Using 1.5 V cells joined in series, how many cells are needed, and how should they be arranged? [Electric Current & its Effects]

- **(A)** 4 cells; in series
- **(B)** 6 cells; in series, all facing the same way
- **(C)** 3 cells; in series
- **(D)** 6 cells; in series with one reversed
- **(E)** 12 cells; in series
- **(F)** 9 cells; in series
- **(G)** 6 cells; in parallel

80. A city's population is 24,76,318. Round it to the nearest lakh. [Large Numbers]

- **(A)** 25,00,000
- **(B)** 24,76,300
- **(C)** 30,00,000
- **(D)** 24,76,000
- **(E)** 24,00,000
- **(F)** 24,80,000

81. A plant pulls up 600 mL of water a day; about 1/100 of it is used to make food and the rest leaves as vapour. How much water is used for food-making, and the tube carrying water up is called? [Nutrition in Plants]

- **(A)** 6 mL; the tube is the phloem
- **(B)** 16 mL; the tube is the xylem
- **(C)** 6 mL; the tube is a stoma
- **(D)** 594 mL; the tube is the xylem
- **(E)** 6 mL; the tube is the xylem
- **(F)** 6 mL; the tube is the chloroplast

82. A transversal cuts two parallel lines. An exterior angle is 130 degrees. Find the interior angle on the same side of the transversal that is next to (linear pair with) its corresponding interior angle. [Lines & Angles]

- **(A)** 50 degrees
- **(B)** 90 degrees
- **(C)** 130 degrees
- **(D)** 40 degrees
- **(E)** 110 degrees
- **(F)** 60 degrees

83. We are told never to touch electrical switches with wet hands. What is the reason, and which material is therefore used to cover switch plates? [Electric Current & its Effects]

- **(A)** Wet skin conducts electricity and can give a shock; switch plates use plastic
- **(B)** Water cools the wire; switch plates use iron
- **(C)** Water carries no current; switch plates use rubber
- **(D)** Wet skin conducts and can give a shock; switch plates use metal
- **(E)** Wet hands make the bulb brighter; switch plates use copper
- **(F)** Wet hands stop the current; switch plates use plastic
- **(G)** Wet skin insulates electricity; switch plates use glass

84. How many whole numbers lie strictly between 4,99,995 and 5,00,005? [Large Numbers]

- **(A)** 4
- **(B)** 5
- **(C)** 10
- **(D)** 9
- **(E)** 8
- **(F)** 11

85. Which word sentence correctly shows photosynthesis, and what is the single source of energy for it? [Nutrition in Plants]

- **(A)** Carbon dioxide + water gives glucose + oxygen; the energy source is the soil
- **(B)** Oxygen + water gives glucose + carbon dioxide; the energy source is sunlight
- **(C)** Carbon dioxide + water gives starch + nitrogen; the energy source is sunlight
- **(D)** Glucose + water gives carbon dioxide + oxygen; the energy source is sunlight
- **(E)** Glucose + oxygen gives carbon dioxide + water; the energy source is sunlight
- **(F)** Carbon dioxide + water gives glucose + oxygen; the energy source is sunlight

86. An angle is exactly one-fifth of a complete turn (a full circle). How many degrees is it, and what type of angle is it? [Lines & Angles]

- **(A)** 36 degrees, an acute angle
- **(B)** 72 degrees, an acute angle
- **(C)** 72 degrees, a right angle
- **(D)** 72 degrees, an obtuse angle
- **(E)** 60 degrees, an acute angle
- **(F)** 45 degrees, an acute angle
- **(G)** 75 degrees, an acute angle

87. An incandescent bulb lights up because current heats its thin filament until it glows white-hot. The filament is made of a metal with a very high melting point - which one, and which effect of current is at work? [Electric Current & its Effects]

- **(A)** Copper; the heating effect of current
- **(B)** Nichrome; the chemical effect of current
- **(C)** Tungsten; the magnetic effect of current
- **(D)** Aluminium; the heating effect of current
- **(E)** Tungsten; the lighting effect only
- **(F)** Tungsten; the heating effect of current
- **(G)** Tungsten; the cooling effect of current

88. A pump moves 1,250 litres of water every hour and runs 6 hours a day. How much water does it move in 3 days? [Large Numbers]

- **(A)** 3,750 litres
- **(B)** 20,500 litres
- **(C)** 2,25,000 litres
- **(D)** 7,500 litres
- **(E)** 2,250 litres
- **(F)** 22,500 litres

89. Pitcher plants and sundews live in swampy soil poor in one nutrient, so they eat insects. Which nutrient is missing, and if a sundew catches 3 insects on Monday and 2 more each following day, how many in all by Wednesday? [Nutrition in Plants]

- **(A)** Nitrogen; 12 insects in all by Wednesday
- **(B)** Phosphorus; 15 insects in all by Wednesday
- **(C)** Nitrogen; 10 insects in all by Wednesday
- **(D)** Nitrogen; 15 insects in all by Wednesday
- **(E)** Oxygen; 15 insects in all by Wednesday
- **(F)** Nitrogen; 7 insects in all by Wednesday

90. Two angles are supplementary and one of them is a right angle. What must the other angle be, and what type is it? [Lines & Angles]

- **(A)** 90 degrees, an obtuse angle
- **(B)** 90 degrees, a right angle
- **(C)** 0 degrees, no angle at all
- **(D)** 45 degrees, an acute angle
- **(E)** 180 degrees, a straight angle
- **(F)** 90 degrees, an acute angle

91. In a simple series circuit with two bulbs, a student says the bulb nearer the battery gets more current. Is this right, and what is true about current in a series circuit? [Electric Current & its Effects]

- **(A)** No; the current doubles at each bulb
- **(B)** No; the current is zero in a series circuit
- **(C)** Yes; current is used up as it goes along
- **(D)** No; the last bulb gets more current
- **(E)** Yes; the first bulb gets more current
- **(F)** No; the current is the same everywhere in a series circuit
- **(G)** Yes; only the near bulb glows

92. Estimate 6,732 - 2,488 by rounding each number to the nearest ten, then give the exact difference. [Large Numbers]

- **(A)** Estimate 9,220; exact 4,244
- **(B)** Estimate 4,200; exact 4,244
- **(C)** Estimate 4,240; exact 4,240
- **(D)** Estimate 4,000; exact 4,244
- **(E)** Estimate 4,240; exact 4,244
- **(F)** Estimate 4,250; exact 4,244

93. A 50 kg bag of fertiliser is 20% nitrogen by mass. How many kg of nitrogen does it hold, and why must such nutrients be added to fields again and again? [Nutrition in Plants]

- **(A)** 10 kg; because the soil makes its own nitrogen
- **(B)** 5 kg; because crops use up nitrogen
- **(C)** 2.5 kg; because crops use up nitrogen
- **(D)** 10 kg; because rain alone adds nitrogen
- **(E)** 10 kg; because crops keep using up the soil's nitrogen
- **(F)** 20 kg; because crops use up nitrogen

94. Two angles forming a linear pair are 130 degrees and 50 degrees. Each is cut in half by a ray. What is the angle between the two bisecting rays? [Lines & Angles]

- **(A)** 40 degrees
- **(B)** 45 degrees
- **(C)** 65 degrees
- **(D)** 100 degrees
- **(E)** 90 degrees
- **(F)** 25 degrees
- **(G)** 180 degrees

95. A geyser takes 8 minutes to heat one tank of water using the heating effect of current. Heating three tanks one after another takes how long, and the longer it runs the - [Electric Current & its Effects]

- **(A)** 11 minutes; the more electricity it uses
- **(B)** 24 minutes; the more heat (and electricity) it uses
- **(C)** 83 minutes; the more electricity it uses
- **(D)** 24 minutes; the brighter it glows
- **(E)** 24 minutes; the less electricity it uses
- **(F)** 16 minutes; the more electricity it uses
- **(G)** 24 minutes; the cooler it gets

96. In 5,234,678 (International system), the digit 2 stands for what value, and which place is it in? [Large Numbers]

- **(A)** 200,000; the millions place
- **(B)** 20,000; the hundred-thousands place
- **(C)** 2,000,000; the millions place
- **(D)** 20,000; the ten-thousands place
- **(E)** 200,000; the hundred-thousands place
- **(F)** 200; the hundreds place
- **(G)** 2,000; the thousands place

97. At night a plant only respires, taking in oxygen. By day it does both, but photosynthesis is much faster. Over a full day, which gas does a green plant give out overall, and which process makes its food? [Nutrition in Plants]

- **(A)** Overall it gives out nitrogen; photosynthesis makes its food
- **(B)** Overall it gives out carbon dioxide; respiration makes its food
- **(C)** Overall it gives out oxygen; photosynthesis makes its food
- **(D)** It gives out no gas; photosynthesis makes its food
- **(E)** Overall it gives out oxygen; transpiration makes its food
- **(F)** Overall it gives out carbon dioxide; photosynthesis makes its food
- **(G)** Overall it gives out oxygen; respiration makes its food

98. On two parallel lines a transversal makes co-interior angles (2x + 20) degrees and (3x + 10) degrees. Find x and both angles. [Lines & Angles]

- **(A)** x = 34, angles 88 degrees and 112 degrees
- **(B)** x = 30, angles 70 degrees and 110 degrees
- **(C)** x = 30, angles 80 degrees and 100 degrees
- **(D)** x = 20, angles 60 degrees and 70 degrees
- **(E)** x = 26, angles 72 degrees and 88 degrees
- **(F)** x = 30, angles 80 degrees and 90 degrees
- **(G)** x = 30, angles 100 degrees and 100 degrees

99. An electromagnet is made stronger in two ways at once: using more cells and more turns of wire. A student goes from 2 cells to 4 cells AND from 40 turns to 80 turns. Both changes do what, and the core is best made of? [Electric Current & its Effects]

- **(A)** Both make it stronger; the core is best made of soft iron
- **(B)** Both make it stronger; the core is best made of steel (a permanent magnet)
- **(C)** More cells help but more turns weaken it; iron core
- **(D)** Both make it weaker; the core is best made of soft iron
- **(E)** Both make it stronger; the core is best made of plastic
- **(F)** Both changes have no effect; iron core

100. A relief fund collects 3,75,000 rupees from one town and 4,86,500 rupees from another, then spends 6,50,000 rupees on aid. How much money is left? [Large Numbers]

- **(A)** Rs 2,01,500
- **(B)** Rs 2,21,500
- **(C)** Rs 3,11,500
- **(D)** Rs 1,11,500
- **(E)** Rs 8,61,500
- **(F)** Rs 2,11,000
- **(G)** Rs 2,11,500